

# Analysis of the ABC Centering Algorithm

## Programmable Matter & Distributed Algorithms

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### Question 1: Is D the center of mass of the set of robots? Explain or prove it!

**Answer:** No, D is not the physical center of mass. It represents the topological center of the network.

### Proof and Explanation

The physical center of mass ( $C_m$ ) of a discrete set of  $N$  robots with uniform mass is defined by the arithmetic mean of their spatial coordinates:

$$C_m = \frac{1}{N} \sum_{i=1}^N \vec{r}_i$$

where  $\vec{r}_i$  represents the 3D position  $(x_i, y_i, z_i)$  of the  $i$ -th robot. The center of mass is heavily influenced by the spatial density and clustering of the modules.

In contrast, the ABC algorithm identifies the topological center (D) by calculating the minimum network eccentricity based strictly on communication hops. The algorithm finds the topological midpoint of the longest path between two extremity modules (B and C):

$$\text{targetCenterDist} = \left\lfloor \frac{\text{dist}(B, C)}{2} \right\rfloor$$

We can prove they are fundamentally different by counterexample:

Consider a barbell-shaped configuration consisting of a dense cluster robots on the left, connected by a thin, linear chain of robots to a small cluster of 2 robots on the right. The physical center of mass ( $C_m$ ) will shift significantly to the left due to the mass of the robot cluster. However, the ABC algorithm only evaluates the communication distance between the extremities, entirely ignoring the density of the clusters. Thus, module D will locate itself near the topological midpoint of the connecting chain, far from the true physical center of mass.

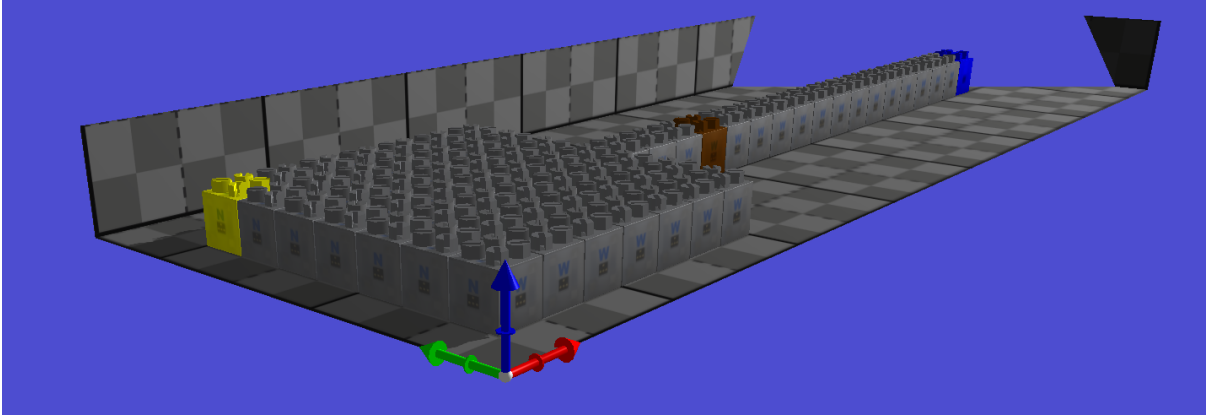


Figure 1: Barbell config proof.

## Question 2: Is D unique in a configuration? Explain or prove it!

**Answer:** No, the topological center D is not inherently unique in a given configuration.

### Proof and Explanation

A modular robot configuration can possess multiple valid topological centers depending on its symmetry and the parity of its longest path. The uniqueness of D in a distributed simulation is typically an artifact of arbitrary tie-breaking mechanisms implemented in the code.

We can prove the lack of absolute uniqueness by counterexample:

1. **Even-Length Paths (Odd Number of Hops):** Consider a simple linear chain of 6 robots connected in series:  $R_1 - R_2 - R_3 - R_4 - R_5 - R_6$ . The extremities are  $B = R_1$  and  $C = R_6$ , with a maximum path distance of 5 hops. The topological midpoint lies exactly halfway between  $R_3$  and  $R_4$ . Both  $R_3$  (distance 2 from B, 3 from C) and  $R_4$  (distance 3 from B, 2 from C) share the exact same minimal network eccentricity. Therefore, both are equally valid topological centers. The ABC algorithm forces a single choice by truncating the math (e.g., integer division  $\lfloor 5/2 \rfloor = 2$ ), but topologically, D is not unique.
2. **Symmetrical Extremities:** In a highly symmetrical topology, such as a cross (+) or a perfect square grid, there are multiple equivalent longest paths. During the Phase 2 flood from extremity B, multiple leaf modules might tie for the maximum distance, meaning there are multiple valid candidates for extremity C. Depending on which C is arbitrarily chosen by the convergecast tie-breaker (e.g., the first acknowledgement received), the resulting routing path, and therefore the elected midpoint D, could fall on entirely different modules.